

Ariel Szekely

Contact Information

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Education

- 2020-2026 **Ph.D. in Computer Science**, *Massachusetts Institute of Technology*, Cambridge, MA, Thesis: *SigmaOS: a Cloud OS with a Unified API for Serverless and Microservice Applications*
Advisors: Frans Kaashoek and Robert Morris
- 2020-2022 **M.S. in Computer Science**, *Massachusetts Institute of Technology*, Cambridge, MA, Thesis: *σ OS: Elastic Realms for Multi-Tenant Cloud Computing*
GPA: 5.0/5.0
- 2016-2020 **B.S. in Computer Science (Honors)**, *The University of Texas at Austin*, Austin, TX, Thesis: *Exploring GPU Timing Side-Channels*
GPA: 3.94/4.0

Interests

Distributed Systems, Operating Systems, Security, Concurrency.

Professional Experience

- 2023 **Student Researcher**, *Google Systems Research Group*, Seattle, WA
Designed and evaluated a system for TPU disaggregation for Ads model serving.
- 2019 **Research Assistant**, *The University of Texas at Austin*, Austin, TX
Developed the Telekine GPU timing side-channel attack for our NSDI '20 paper.
- 2018 **Software Engineer Intern**, *Bloomberg L.P.*, New York City, NY
Back-end development.
- 2017 **Software Engineer Intern**, *AT&T*, Dallas, TX
Full-stack development.

Publications

- arXiv pre-print **Fast end-to-end cloud application cold-start with initscripts**, Ariel Szekely, Hannah Gross, Robert Morris, Frans Kaashoek
Designed and implemented initscripts, a new scriptable interface for cloud developers to specify their application's initialization routine to the provider in order to overlap application initialization with infrastructure setup costs. Initscripts reduce cold-start latency of serverless applications by 2.6-1.8X and microservices like etcd and memcached by 1.7X.

- SOSP '24 **Unifying serverless and microservice workloads with SigmaOS**, Ariel Szekely, Adam Belay, Robert Morris, M. Frans Kaashoek
Designed and implemented SigmaOS, a novel cloud operating system and cluster manager which supports both microservices and serverless applications. SigmaOS's API enables applications to warm-start in under 2ms and cold-start in under 8ms with strong isolation in a lightweight container. [GitHub]
- ASPLOS '23 **Towards a machine learning-assisted kernel with LAKE**, Henrique Fingler, Isha Tarte, Hangchen Yu, **Ariel Szekely**, Bodun Hu, Aditya Akella, Christopher J. Rossbach
Rewrote the Linux Kernel Samepage Merging (KSM) module to benefit from hardware acceleration.
- NSDI '20 **Telekine: Secure computing with cloud GPUs**, Tyler Hunt, Zhipeng Jia, Vance Miller, **Ariel Szekely**, Yige Hu, Christopher J. Rossbach, Emmett Witchel
Built a timing side-channel attack against a GPU Trusted Execution Environment (TEE), allowing the attacker to learn the class of images passed through ResNet by timing the GPU kernels.

Awards

- 2022 NSF Graduate Research Fellowship
- 2020 MIT Lemelson Presidential Fellowship
- 2020 MIT EECS Departmental Fellowship
- 2020 Dean's Honored Graduate, UT Austin College of Natural Sciences
- 2020 Research Distinction, UT Austin College of Natural Sciences
- 2020 NSF Graduate Research Fellowship Honorable Mention
- 2020 CRA Outstanding Undergraduate Researcher Honorable Mention
- 2019 NSF REU
- 2019 UT Endowed Presidential Scholarship
- 2018,19,20 UT College of Natural Sciences College Scholar
- 2018 Louis E. Rosier Memorial Scholarship
- 2016 UT CNS Freshman Scholarship
- 2016 National Merit Scholarship

Mentorship and Advising

- 2026 **Nicolas Camenisch**, MIT, M.Eng. Thesis
 Thesis title: *Fork Around and Find Out: Accelerating Dynamic Language Runtimes in SigmaOS via Lightweight Container Forking*
- 2026 **Nathan Mustafa**, MIT, M.Eng. Thesis
 Thesis title: *Efficient RDMA for SigmaOS*
- 2025 **Ivy Wu**, MIT, M.Eng. Thesis
 Thesis title: *Interposing the syscall boundary: Transparent python execution in SigmaOS*
- 2025 **Ryan Chang**, MIT, M.Eng. Thesis
 Thesis title: *Optimizing SigmaOS for efficient orchestration of fault-tolerant, burst-parallel workloads*
- 2025 **Freddie Tang**, MIT, M.Eng. Thesis
 Thesis title: *Accelerating Burst Parallelism of SigmaOS processes with CRIU*
- 2025 **Katie Liu**, MIT, M.Eng. Thesis
 Thesis title: *Enabling efficient ML inference in SigmaOS with model-aware scheduling*
- 2024 **Kevin S. Chen**, MIT, M.Eng. Thesis
 Thesis title: *IC σ OS: σ OS for Intercloud Environments*

2023 **Yizheng He, MIT, M.Sc. Thesis**

Thesis title: *Evaluating sigmaos with kubernetes for orchestrating microservice and serverless applications*

Service

2021 **SOSP '21 Artifact Evaluation Committee**, *SOSP 2021*, Kolbenz, Germany

2022-present **EECS Communication Lab Fellow**, *Massachusetts Institute of Technology*, Cambridge, MA

2020-2022 **MIT Graduate Application Assistance Program Mentor**, *Massachusetts Institute of Technology*, Cambridge, MA

Teaching Experience

2023 **TA: Computer Systems Security**, *Massachusetts Institute of Technology*

Instructors: Nickolai Zeldovich

2022 **TA: Operating Systems Engineering**, *Massachusetts Institute of Technology*

Instructors: Frans Kaashoek, Robert Morris

2020 **TA: Concurrency (Honors)**, *The University of Texas at Austin*

Instructor: Christopher J. Rossbach

2020 **TA: Intro to CS Research (Honors)**, *The University of Texas at Austin*

Instructor: Calvin Lin

2019 **TA: Intro to CS Research (Honors)**, *The University of Texas at Austin*

Instructor: Calvin Lin

Languages

English Native

Spanish Native

French Fluent